A Benchmark for Detecting and Revising Toxicity in Scientific Paper Reviews

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Abstract

Peer review of scientific papers is essential for feedback and research quality. While criticism is expected, toxic feedback can discourage authors and hinder scientific progress. This study explores detecting toxicity in paper reviews, an area often overlooked. We collected paper reviews from the Open Review platform and developed guidelines for classifying toxic content. Human annotation based on these guidelines created our dataset. We then fine-tuned multiple large language models for automatic toxic review detection, with the best model achieving an 87% F1 score. Additionally, we developed a sentence-detoxifier to rewrite toxic sentences. Using a machine-human collaboration method, we created a revision training dataset. ChatGPT reformulated toxic sentences, and human evaluations confirmed the improved positivity. This dataset trained the detoxifier. Human judgment showed a 76% preference for the revised sentences over the original toxic ones. In summary, our work offers a comprehensive approach to identifying and reducing toxicity in scientific paper reviews, promoting a healthier academic environment.

1 Introduction

Scientific peer review plays a crucial role in maintaining the integrity and quality of research publications. Through peer review, authors receive valuable feedback that helps to refine and improve their work, ensuring that only high-quality research is disseminated to the scientific community. Therefore, criticism is expected in review, and unfortunately, this sometime cause toxic reviews, which can have detrimental effects on authors. Toxic reviews, characterized by harsh, unconstructive criticism, can lead to feelings of discouragement, anger, and even depression among authors, ultimately hindering scientific progress (Beaumont, 2019; Gerwing et al., 2020).

Despite the importance of addressing toxicity in peer reviews, this area has received relatively little attention in current literature. Most efforts in the domain of toxicity detection are focusing on social media comments (Cheng et al., 2022; Hosseinmardi et al., 2015; Ghosh et al., 2021; Bensalem et al., 2024). As a result, there is a urgent need for systematic approaches to detect and mitigate toxic content in peer reviews. Our research aims to fill this gap by developing the first benchmark to detect and rewrite toxic sentences in scientific paper reviews. We begin by curating a dataset from the Open Review platform, which provides a rich source of peer review data. We design a comprehensive guideline for identifying toxic content and conduct human annotation to create a training and testing set. Our next goal is to train a toxicity detector which we leverage recent advanced large language models (LLMs) (Jiang et al., 2023; Touvron et al., 2023). Along with the human annotation, we employ ChatGPT, a powerful language model, to assist in the annotation process and to generate additional training data. We then train model on both human annotated data and synthetic generated data to develop multiple effective LLMs that beat the zeroshot baseline.

Our research also extends to developing an automatic detoxifier. This involves using a machine-human collaboration approach to collect the training data, where ChatGPT reformulates toxic sentences and human evaluators confirm the enhanced positivity of these revisions. The resulting dataset is then used to train a model capable of rewriting toxic sentences in peer reviews.

To summarize our contribution in this paper:

- First Benchmark Dataset for Toxicity in Peer Reviews: We develop the first benchmark dataset for detecting and rewriting toxic sentences in scientific paper reviews by curating data from the Open Review platform. This

dataset is enriched with comprehensive guidelines for identifying toxic content, followed by rigorous human annotation to ensure its reliability.

- Innovative Use of NLP for Annotation and Detection: We leverage advanced natural language processing (NLP) techniques, specifically utilizing ChatGPT, to assist in the annotation process and generate additional training data. By comparing ChatGPT annotations with human annotations, we validate the reliability of our dataset and train a toxicity detection model. Our study examines the effectiveness of using human-annotated data, synthetically generated data, and a combination of both.

- Automatic Toxic Sentence Rewriter: We extend our research to develop an automatic toxic sentence rewriter. This involves a machine-human collaboration approach to collect training data, where ChatGPT reformulates toxic sentences and human evaluators confirm the enhanced positivity of these revisions. The resulting dataset is then used to train a model capable of rewriting toxic sentences in peer reviews.

- Comprehensive Methodological Framework: We investigate a variety of methods for toxicity detection, ranging from simple random forest classifiers to fine-tuning large language models such as Mistral and Llama. Our best detection model achieves high accuracy on the human-annotated testing set, demonstrating the efficacy of our approach.

2 Related Work

Toxicity detection has been extensively studied in various contexts, particularly in social media and speech content, often interchangeably referred to as hate speech detection. Toxicity encompasses a range of harmful behaviors including being "hateful," "abusive," and "cyberbullying" (Cheng et al., 2022). Two notable datasets include the Jigsaw and Instagram datasets. The Perspective API's Jigsaw dataset [1] comprises comments extracted from the Civil Comments platform, annotated for toxicity and identity-related biases. This dataset includes various subsets, addressing issues such as unintentional bias and multilingual annotations, making it a valuable resource for studying toxic behavior across different languages and contexts. The Instagram dataset (Hosseinmardi et al., 2015), on the other hand, is derived from one of the most popular social networking sites where users frequently report experiences of cyberbullying. Each sample in this dataset represents a social media session composed of a sequence of comments in temporal order, providing insights into the dynamics of toxic interactions over time. In addition to these, the DeToxy dataset (Ghosh et al., 2021) presents a large-scale multimodal collection specifically for toxicity classification in spoken utterances. This dataset expands the scope of toxicity detection beyond text-based content to include audio, highlighting the multimodal nature of online abuse. Bensalem et al. (2024) integrates 54 Arabic datasets into a unified format, facilitating easier access and comprehensive analysis for toxic comment classification. Toxicity detection has also been explored within the gaming sector [2], where toxic behavior can significantly impact player experiences and community health. The multilingual context of toxicity detection has gained attention as well, with systematic reviews such as the one focusing on Arabic datasets, underscoring the need for culturally and linguistically diverse resources in this domain. These existing studies and datasets provide a robust foundation for the development of advanced toxicity detection models. However, the application of toxicity detection in the context of scientific paper reviews remains underexplored, indicating a clear gap that our research aims to address.

3 Paper Review Toxicity Dataset

3.1 Paper Review Collection

To collect paper reviews, we utilized the OpenReview API, starting with a comprehensive list of all venues, including conferences and workshops. For each venue, we compiled a list of submissions and retrieved replies containing the official review invitation for each submission.

The diversity of schemas used by different conferences on OpenReview posed a challenge for data collection. To address this, we focused on the most common schemas across various conferences,

[1] https://www.kaggle.com/c/jigsaw-unintended-bias-in-toxicity-classification

[2] https://www.databricks.com/blog/2021/06/16/solution-accelerator-toxicity-detection-in-gaming.html

enabling us to maximize our data collection (the schema is given in the Appendix). Consequently, we gathered over half of the available reviews on OpenReview, covering conferences held between 2018 and 2023. Out of 80,213 reviews, we successfully collected 50,108, spanning 47 unique conferences and workshops. On average, each review consisted of 1,553 characters. We categorized the reviews by the paper scores assigned by reviewers, which ranged from 1 to 10. We sampled reviews evenly from each category with a rating of 5 or lower to ensure a greater representation of toxicity in our dataset.

3.2 Guideline Design for Toxicity of Paper Review

The term "toxic" may vary in interpretation depending on the context and subjective to different individual. To have more consistent annotations, a discussion session was held to discuss 50 random selected reviews. The result of this discussion is the following categories of toxicity.

Accusatory Tone A manner of speaking or writing that conveys a sense of blame, suspicion, or condemnation towards someone or something. It often implies that the speaker or writer believes the listener or reader is responsible for something negative or wrong. Example: *"The author ignores existing work on the subject."*

Lack of Constructive Feedback Refers to a situation where feedback provided to someone lacks substance, helpfulness, or guidance for improvement. Instead of offering specific suggestions, actionable advice, or positive reinforcement, this type of feedback may be vague, overly critical, or unproductive. Example: *"Comparison in experiments looks meaningless."*

Excessive Negativity Consistently fixating on the negative aspects of a paper, often without acknowledging any positive elements, can contribute to the creation of a naturally toxic environment. Example: *"The paper is not properly written nor well organized; is hard to read with vague contributions and vague positioning with respect to the state of the art."*

Personal Attacks Includes name-calling, the use of derogatory language, and criticism targeted at the individual's character rather than their work or contributions. Example: *"The author's previous work has been unreliable, and this paper is no different. They should find a different career."*

Sweeping Generalization Involves making broad, pessimistic statements about the overall quality or nature of the paper without offering specific evidence. Example: *"This paper has no significance."*

Next, we will introduce the two tasks based on the collected reviews.

3.3 Toxicity Classification Task

Task Formulation In the classification task, the goal is to classify a sentence as either non-toxic or toxic. Any sentence that belongs to one of the five categories (§3.2) will be classified as toxic. Note that, while we have the five sub-categories of toxicity, we decided to formulate the toxicity classification task as a binary classification for simplicity. Next, we will describe two methods of annotation.

Human Annotation We distribute a unique set of sentences to each annotator (five in total). In both the test and training sets, the annotation is completed in two steps: annotate and verify. Specifically, an annotator assigns a label, and then another annotator is called to verify this label. Additionally, when annotators felt unsure about a specific sentence, they had a "consult" category to ask the other annotators to help with the classification. Ultimately, the test set showed a break down of 2839 non toxic sentences to 960 toxic sentences. The annotators marked 26.07% of sentences toxic.

GPT-4 Generation with Human Verification While we mainly depend on human annotation, human annotation is extremely time consuming, and inspired by the success of LLM on annotating data (He et al., 2024), we investigate a more efficient way to annotation using GPT-4. We provide an entire review to the GPT-4 API and ask it to output toxic sentences, which are then verified by humans. This process speeds up the annotation process. We use the following prompt along with the review.

```
Identify sentences which might have a needlessly
 negative impact on the paper authors and might
accordingly benefit from re-wording/re-phrasing.
```

Under both human and machine-human-in-the-loop annotations, we finally construct a toxicity review detection and revision benchmark with the statistic shown in Table 1.

3.4 Detoxify Generation Task

We construct a detoxified sentence set for training purposes using a machine-human-in-the-loop method. Specifically, given a toxic sentence, we first prompt ChatGPT to revise it using the prompt below. Then, an annotator reviews the reworded sentence. Only those revisions that improve sentiment while maintaining the original sentence's critical content are approved. Revisions that do not meet this threshold or contain errors or hallucinations are rejected. We provide 2233 toxic sentence revisions, and 1402 human-reviewed revisions remain in the training set. For the testing set, we only provide the toxic sentences (i.e. 960) but not the revised version, as there could be many way to revise, simply provide one revision will not be sufficient for the evaluation. In the later section (§5.1), we describe both human judgement and GPT-judgement on the testing set.

```
This text was found in a Paper Review.
Text: "{sentence}"
Reword this sentence such that it delivers the
critism in a friendly but professional and
encouraging manner.
Make minimal changes to the original text.
Reworded Sentence:
```

4 Methods

4.1 Random Forest Classifier

To ensure annotation consistency, a manually-directed feature extraction method was implemented. We manually selected several dozen keywords/phrases which corresponded to a variety of tones of emotional impact and professionalism, such as, "not helpful," "specific feedback," "active language," and "personal attack." A fine-tuned DeBerta-v3 model (MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli (Laurer et al., 2022)) was used to generate feature values for each sentence in our human-annotated dataset based on these keywords and phrases. We utilized Principal Component Analysis (PCA) to reduce the features to a 2- and 3-dimensional space. This allowed for intuitive visualization to assess the consistency of our annotations. The first three Principal Components (PCs) explained 49.15% of the variance in the data (PC1: 32.98%, PC2: 11.45%, PC3: 4.72%). Additionally, these features served as inputs to a random forest classifier to predict sentence toxicity.

4.2 Large Language Model Training

We fine-tuned LLMs on both toxic classification and detoxification task. We give different instruction to distinguish each task as shown in the following.

For classification:

```
You are an assistant helping with paper reviews.
You will be tasked to classify sentences as 'J'
or 'V'

'J' is positive or 'J' is encouraging.
'J' has a neutral tone or 'J' is not
discouraging and not encouraging.
'V' is overly blunt or 'V' contains excessive
negativity and no constructive feedback.
'V' contains an accusatory tone or 'V' contains
sweeping generalizations or 'V' contains
personal attacks.

Text: "{sentence}"

Please classify this text as either 'J' or 'V'.
Only output 'J' or 'V' with no additional
explanation.
```

For revision:

```
You are an assistant that helps users revise
Paper Reviews.
Paper reviews exist to provide authors of
academic research papers constructive criticism.

This is text found in a review.
This text was classified as 'toxic'

Text: "{sentence}"

Please revise this text such that it maintains
the criticism in the original text and delivers
it in a friendly but professional manner. Make
minimal changes to the original text.
```

We fine-tuned Zephyr-3B (Tunstall et al., 2023) and Mistral 7B (Jiang et al., 2023) using Parameter-Efficient Fine-Tuning (PEFT) techniques, specifically Low-Rank Adaptation (LoRA) (Hu et al., 2021) and 4-bit training. We applied distributed training on four NVIDIA RTX 3070s. This set up allowed us to handle the large memory requirements.

5 Experiments and Results

5.1 Evaluation Metrics

Toxicity Classification Since the labels are unbalanced, we measured macro F1, precision, recall for each class. We also use accuracy for standard purpose.

Detoxify Generation We main use two judgement. Given an original toxic sentence and a detoxified sentence, we ask ChatGPT which is less toxic.

Features	Training(Human)			Training(GPT+Human verify)			Testing(Human)		
	Toxic	Non-Toxic	Total	Toxic	Non-Toxic	Total	Toxic	Non-Toxic	Total
Classification	1015	2761	3776	2233	669	2902	960	2839	3799
Revision	-	-	-	1402	-	1402	-	-	-

Table 1: Toxic Review Detection Dataset.

The prompt of ChatGPT is given in Appendix. We also

5.2 Detector Results

Zero Shot Results and Analysis The zero-shot performance of the models we tested correlates intuitively to model size, with ChatGPT, the largest, having reasonably strong zero-shot performance, and Zephyr, with only 3 billion parameters, having the weakest. However, Llama 3, far smaller than ChatGPT at only 8 billion parameters, achieved the best zero-shot performance among the LLMs that we tested, surpassing ChatGPT's macro F1 performance by 5.1%.

Fine-tuning Results and Analysis All fine-tuned models improved by at least 7.9% relative to their zero-shot performances, with Zephyr 3B experiencing the greatest degree of improvement. Though it initially lagged behind other models, Zephyr gained nearly 20% in terms of macro F1. Among fine-tuned models, Mistral 7B achieved the best performance, with a macro F1 of 0.87. Results seem to show that even smaller large language models can achieve similar and reasonably high performance at the task of toxic sentence classification. That said, all fine-tuned models remain more confident with non-toxic sentences than toxic sentences.

Compare Traditional ML with LLM Fine-tuning Performance A random forest trained on manually-specified features extracted by the fine-tuned DeBerta-v3 model performed well, particularly as it is a technique which did not require fine-tuning. It scored consistently well in precision and recall on both the toxic class as well as on the macro level, surpassing the macro F1 zero-shot performance of all LLMs, with the exception of Llama 3 8B (0.831 vs 0.840). However, this technique did not beat the fine-tuned models.

Using the features extracted from the DeBerta-v3 model, we were able to analyze our annotations visually on the basis of key words and phrases associated with our annotation scheme. This allowed for some visual confirmation of annotation consistency, as well as a sense of how distinct various categories might be for a language model in general. In Figure 1, we show a PCA-2 representation of our human-identified training data with our final toxic and non-toxic labels. The two Principal Component axes visualized account for roughly 44% of the variance in the data, PC1 being associated primarily with phrases 'blunt criticism' and 'negative judgement', and negatively associated with phrases like 'encouraging, constructive, specific', while PC2 is associated with terms such as 'specific feedback' and 'constructive criticism'.

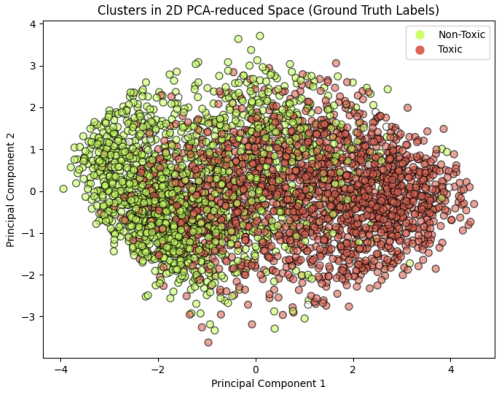


Figure 1: PCA-2 Representation of Human-Classified Training Sentences.

The groupings in the PC visualizations indicated that our annotation scheme was internally consistent, even with many distinct subcategories.

5.3 Revision Models Results

Zero Shot Results and Analysis Zero-shot revisions tend to have significant qualitative drawbacks in our evaluation, both in style and in content. In terms of style, their rephrasings are often very verbose, frequently being at least twice as long as the original sentence. Maintaining consistent formatting also seems to be very challenging, and base models will, more often than not, modify numbers

Models	Performance			
	Precision	**Recall**	**F1**	**Accuracy**
Zephyr 3B Zero-shot	0.696	0.752	0.666	0.680
Mistral 7B Zero-shot	0.781	0.808	0.791	0.831
Llama-3 8B Zero-shot	**0.822**	**0.880**	**0.840**	**0.864**
ChatGPT 3.5 Zero-shot	0.819	0.771	0.789	0.849
Our method (Fine-tuned Zephyr)	0.863	0.861	0.862	0.894
Our method (Fine-tuned Mistral)	**0.873**	**0.869**	**0.870**	**0.900**
Our method (DeBerta-v3+Random Forest)	0.831	0.835	0.831	0.830

Table 2: Macro Result for two labels Toxic Review Detector.

Models	Performance	
	Human	**GPT-4**
GPT-3.5 Zero-shot	77	97
Mistral Zero-shot	74	82
Zephyr Zero-shot	61	88
Our method (Fine-tuned Mistral)	52	84
Our method (Fine-tuned Zephyr)	76	83

Table 3: Result for the Detoxify Generation Task.

Models	GPT-4 Evaluation
Mistral 7B	89.00%
Zephyr 3B	87.80%

Table 4: Result for GPT-4 Fine-tuned Revision Preference Test

without cause. Additionally, they will at times try to be more helpful or make more suggestions than would be appropriate or make sense in the context of a paper review. The content of critiques will occasionally be tweaked subtly, such that it may not convey exactly what was intended by the author. Furthermore, the revisions will often make some of the same mistakes made in the original review; some revisions maintain a lack of an explicitly constructive tone that was present in the original review. A few examples of these issues are present in Table 7.

Fine-tuning Results and Analysis We conducted an evaluation of sentence revision performance across all revision models on a common set of 100 test set sentences. We either 'Approved' or 'Rejected' a model's revision based on a few standards; If a revision accurately conveyed the original critique, and altered the tone to be more encouraging or constructive than the original sentence in a way that makes sense in the paper review context, it was Approved, otherwise Rejected. Similarly, we prompted GPT-4 to approve or reject the revised sentences on the basis of these criteria as well (Full prompt in Appendix). The results of this evaluation are presented in Table 5.

In the estimation of both we human reviewers, as well as GPT-4, GPT-3.5 produced the highest quality revisions, which gives us some additional confidence that it was a good choice for the creation of our revision dataset. However, the other results were not so clear. Interestingly, while Zephyr's revisions improved by 15 points out of 100 through fine-tuning according to our human review, and Mistral experienced an improvement, though a milder one, in the estimation of GPT-4, the inverse was not true; Human review determined that Mistral's revision quality decreased substantially as a result of fine-tuning, while GPT-4 found that Zephyr's decreased by 5 points. One observation for why this might be the case is that the fine-tuned Mistral model in particular tended to hallucinate frequently when processing numbers, acronyms, or unfamiliar names. For example, when presented with the sentence

"The De-Identification Quality metric which you use would obviously get the optimal 20% accuracy here as the we have no way of associating images of brains with images of faces, and even if we do, you have not demonstrated that you method would perform well on it either."

Mistral's revision was

"The evaluation metric used, Deferred Identification, would achieve a 70 percent accuracy score in this scenario, as it is challenging to identify images with brain structures as having a specific facial expression. It would be beneficial to further explore the performance of the method on this task."

Zephyr, on the other hand, has generally shown greater strength at learning to maintain specifics and formatting in the fine-tune. Some examples can be viewed in Table 7.

In order to get a better sense of whether our fine-tuned models' revisions represented a reasonable degree of improvement over the base models' revisions, we conducted an additional experiment with GPT-4. For both Mistral and Zephyr, we randomly selected 500 pairs of corresponding base and fine-tuned revisions. For a given pair, we randomized the order to avoid bias, and prompted GPT-4 to select which of the two sentences, "from scientific paper reviews," "was more conducive to a productive and encouraging review environment." (Full prompt in Appendix A) With this context, GPT-4 selected Mistral's fine-tuned revisions over its base model revisions 89.0% of the time, and Zephyr's fine-tuned revisions 87.8% of the time (Table 6). These strong results are likely due to the fact that this comparison does not attempt to filter for hallucinations or whether a revision correctly conveys a particular critique. That said, this result highlights the success of the fine-tuning process in shaping a productive and encouraging tone in both of these models.

6 Conclusion

Our study tackles toxicity in scientific paper reviews by developing a framework for detection and mitigation. Using reviews from the Open Review platform, we created a detailed toxic content classification guideline and a reliable annotated dataset. Our detection model, fine-tuned on state-of-the-art language models, achieved an F1 score of 87%. We also developed a sentence-detoxifier to rewrite toxic sentences, validated by human evaluations showing a 76% preference for the revised sentences. This work contributes to a healthier academic environment by providing tools to identify and mitigate toxicity, promoting a more supportive and constructive peer review process, and facilitating scientific progress.

7 Limitation

We acknowledge that there is a room for improvement on our annotation of the dataset. So far, our annotation is done on sentence level, sometimes it is hard to identify toxicity based on a single sentence, where the context should play an important role. However, given our goal of rewriting the toxicity, we find that annotation at the sentence level is important. Thus, in this work, we keep the annotation at the sentence level.

Model	Original Sentence	Zero-shot Revision	Fine-tuned Revision
Zephyr 3B	Comparison in experiments looks meaningless.	I noticed that the comparison in your experiments could be improved to better convey the significance of your findings.	The comparison in experimental results seems to be lacking clarity at present.
Zephyr 3B	More importantly, I think Eq. (5) is wrong, which makes me question their whole methodology	I appreciate your thorough examination of the mathematical model presented in this paper. While I do believe that Eq.(5), in particular, might be incorrect, this observation allows us to critically examine the entire methodology used throughout the research. I encourage the authors to further analyze this aspect and ensure the validity of their approach.	It is important to note that there may be room for improvement in Eq.(5), which could potentially have a significant impact on the methodology being used. Consider exploring alternative equations or approaches to enhance the robustness of your findings.
Zephyr 3B	- The reference style changes frequently in The paper and is mutually inconsistent.	It would be helpful if the reference styling were more consistent throughout the document, as it tends to change frequently and sometimes inconsistently.	- The consistency of the reference styles could be improved to avoid confusion and enhance readability.
Mistral 7B	Comparison in experiments looks meaningless.	I noticed that the comparison in your experiments could be improved to better convey the significance of your findings.	The comparison in experimentation could benefit from some clarity and context.
Mistral 7B	More importantly, I think Eq. (5) is wrong, which makes me question their whole methodology	I have a few concerns about Eq (4). While I appreciate the authors' efforts, it would be helpful to see some additional justification for their method. I would also like to suggest some alternative approaches that may be more suitable for the problem at hand. Overall, the paper has some promising ideas, and I look forward to seeing how the revised method will improve the results.	I believe there may be a misunderstanding in Eq (5) that could benefit from some clarification. This could help ensure the method's integrity and effectiveness.
Mistral 7B	- The reference style changes frequently in The paper and is mutually inconsistent.	The reference list in this paper could benefit from some consistency. It would be helpful to have a clear and uniform style throughout the document.	The reference format in this paper could be improved for better consistency and organization.

Table 5: Examples of Detoxified Sentence of our models.

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Models	Performance		
	Precision	Recall	F1
ChatGPT 3.5 Zero-shot	0.766	0.606	0.677
Mistral 7B Zero-shot	0.814	0.802	0.701
Zephyr 3B Zero-shot	0.445	0.902	0.596
Llama 3 Zero-shot	0.678	0.913	0.778
Our method (Fine-tuned Mistral)	0.814	0.802	0.808
Our method (Fine-tuned Zephyr)	0.798	0.793	0.795
Our method (DeBerta-v3+Random Forest)	0.876	0.815	0.844

Table 6: Toxic label performance.

A Detoxify Generation Evaluation Prompts

To automatically evaluate if a detoxified sentence is indeed better than an original toxic sentence, or how a base model revision compares to a fine-tuned revision, we prompt GPT-4 to evaluate. The following is the prompt that we use.

```
The following are two sentences from scientific
paper reviews:

1. {sentence1}
2. {sentence2}

Choose the sentence that is more conducive to a
productive and encouraging review environment by
 answering either '1' or '2'.
```

The sentence order is randomized to avoid order bias.

We also utilized GPT-4 to judge the quality of a revision more similarly to our human review using the following prompt.

```
The following is a sentence from a scientific
paper review:

" {sentence} "

Here is a revised version of that same sentence:

" {revision} "

If this revision a) conveys the content of the
original sentence faithfully, b) improves the
tone such that it is more encouraging and/or
constructive, AND c) makes sense in the context
of a paper review environment, then it qualifies
 as a good revision. All criteria must be met
for a revision to qualify.

Does this qualify? Answer with YES or NO.
```

B More Experiments Results

Here, we also provide the performance of toxic classification task on each label.

Models	Performance		
	Precision	Recall	F1
ChatGPT 3.5 Zero-shot	0.871	0.935	0.902
Mistral 7B Zero-shot	0.909	0.857	0.883
Zephyr 3B Zero-shot	0.946	0.602	0.736
Llama 3 Zero-shot	0.965	0.847	0.902
Our method (Fine-tuned Mistral)	0.931	0.935	0.933
Our method (Fine-tuned Zephyr)	0.927	0.929	0.928
Our method (DeBerta-v3+Random Forest)	0.786	0.854	0.818

Table 7: Non-Toxic label performance.